Crypto Momentum Timing

Applying the Bookmaker Pool Model to Bitcoin

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This report documents two experiments: a failed pairs trade (BTC/ETH) and a successful timing strategy (hold BTC or cash). All results are from backtests on historical data. Past performance does not guarantee future results.

The Intuition: A Day at the Races

Before the mathematics, consider a horse race.

The Bookmaker's Pool

A bookmaker doesn't set the odds himself — the *punters* do, by where they put their money. Suppose three horses are running in the Melbourne Cup, and the total pool is $100,000:

Horse	Money bet	Share of pool	Implied odds
Thunderbolt	$60,000	60%	1.67 to 1 ($0.67 profit per $1)
Silver Arrow	$30,000	30%	3.33 to 1 ($2.33 profit per $1)
Dark Horse	$10,000	10%	10 to 1 ($9.00 profit per $1)
Total pool	**$100,000**	**100%**	

The odds tell you what the *crowd* thinks. Thunderbolt is the favourite because the most money backs it. But here's the key insight: **the odds keep changing as more money comes in.**

Suppose a rumour spreads that Dark Horse had a secret training session and looked incredible. Suddenly, $20,000 of new money pours in on Dark Horse:

Horse	Money bet	Share of pool	Implied odds
Thunderbolt	$60,000	50%	2.0 to 1
Silver Arrow	$30,000	25%	4.0 to 1
Dark Horse	$30,000	25%	4.0 to 1
Total pool	**$120,000**	**100%**	

Dark Horse's odds have shortened from 10:1 to 4:1. Not because the bookmaker decided it was faster — but because *the money moved*. The pool reflects the collective conviction of everyone with skin in the game.

From Horses to Bitcoin

Now replace the horses with a stock — or Bitcoin. Each trading day, people place their bets:

- **Buyers** put money into the pool (they think the price should be higher).
- **Sellers** take money out (they think the price should be lower).
- The **volume** tells you how much money moved.
- The **daily return** tells you which direction it moved.

On a day when Bitcoin rises 3% on heavy volume, that's like a rush of money pouring in on Thunderbolt — a large, high-conviction bet that the price should be higher. On a day when Bitcoin falls 2% on light volume, that's like a few punters nervously pulling money off Silver Arrow — a small, low-conviction bet that the price should be lower.

The **momentum price** is what the pool says the asset is worth, after accumulating all these daily bets over time. If the pool says "$110,000" but the market price is $95,000, then the weight of money — the collective conviction of everyone who has traded — supports a higher price. The asset is **undervalued by momentum**.

That's when you buy.

If the pool says "$80,000" but the market price is $95,000, the money has been flowing out. The crowd is quietly leaving. The asset is **overvalued by momentum**.

That's when you hold cash and wait.

The Algorithm: Formal Definition

The following makes the bookmaker analogy precise.

Mathematical Definition

Given a time series of daily closing prices P_t and volumes V_t:

1. Compute daily returns: $r_t = \dfrac{P_t - P_{t-1}}{P_{t-1}}$
2. Normalise volume (to prevent overflow on crypto's large numbers): $\hat{V}_t = \dfrac{V_t}{\bar{V}}$ where $\bar{V}$ is the mean volume over the lookback window.
3. Compute net directional flow: $f_t = \hat{V}_t \times r_t$

   On up days ($r_t > 0$), f_t is positive — buyers are winning. On down days ($r_t < 0$), f_t is negative — sellers are winning. On balanced days, the flows cancel out.
4. Update the pool iteratively:

$$\begin{aligned} \text{pool}_0 &= P_0 \\ \text{pool}_t &= \text{pool}_{t-1} + f_t \times \text{pool}_{t-1} \end{aligned}$$

5. The momentum price is the final pool value: $M = \text{pool}_T$

The Signal

The **momentum ratio** is:

$$\rho = \frac{M}{P_T}$$

- $\rho > 1$: The volume-weighted flow of money supports a *higher* price than the market is offering. The asset is **undervalued** by momentum. **Hold the asset.**
- $\rho < 1$: The volume-weighted flow of money supports a *lower* price. The asset is **overvalued** by momentum. **Hold cash.**

The lookback window (how many days of history feed the pool) is a tunable parameter. We tested 90, 180, and 365 days.

Origin

This model was originally developed for ASX 100 equities (~`/git/momentum/`), where it produced +13.6% alpha per 6-month period over 10 consecutive periods ($p = 0.046$). This report tests whether it transfers to cryptocurrency.

Experiment 1: BTC/ETH Pairs Trade (Failed)

Hypothesis

When Bitcoin is over-momentum-priced and Ethereum is under-momentum-priced simultaneously, rotate from BTC into ETH. Rotate back when the spread recalibrates. This is a **relative value** or **pairs trade**.

Result

Metric	90-day	180-day
Strategy return	−91.1%	−42.9%
ETH buy & hold	+311.8%	−21.3%
BTC buy & hold	+305.0%	+19.7%
Sharpe ratio	−0.70	−0.48
Max drawdown	−94.3%	−69.1%
Trades	11	3

Verdict: Total failure. The strategy lost money across every configuration tested.

Why It Failed

1. **Correlated momentum signals.** BTC and ETH are driven by the same macro forces (interest rates, risk appetite, regulatory news). Their momentum ratios move in lockstep. The spread between them is noise, not signal.
2. **Secular trend.** ETH has been in a structural decline against BTC since mid-2022. A mean-reversion strategy on a pair that is *not mean-reverting* is guaranteed to lose.
3. **The original model works on independent assets.** On the ASX, BHP and CSL have genuinely different fundamentals. BTC and ETH do not — they are both "crypto."

Conclusion: The bookmaker model's power comes from detecting divergence between momentum and price in assets with independent fundamentals. When two assets share the same drivers, the signals converge and the pairs trade has no edge. This approach is **abandoned**.

Experiment 2: Single-Asset Timing (Promising)

Hypothesis

Use the momentum ratio as a timing signal on a **single asset**: hold when $\rho > 1$ (undervalued by momentum), go to cash when $\rho \leq 1$ (overvalued). No pairs, no shorting — just timing entries and exits.

Results: Bitcoin (BTC-USD)

Period: 1 January 2021 to 24 August 2026 (2,062 trading days).

Metric	90-day	180-day	365-day
Strategy return	+39.6%	**+462.7%**	+132.3%
Buy & hold return	+167.5%	+167.5%	+167.5%
Alpha	−127.9%	**+295.2%**	−35.2%
Sharpe ratio	0.35	**1.16**	0.66
Max drawdown	−59.5%	**-28.1%**	−33.2%
Buy & hold drawdown	−76.6%	−76.6%	−76.6%
Days in BTC	34.0%	34.3%	30.2%
Days in cash	66.0%	65.7%	69.8%
Distinct trades	22	16	10
Win rate	45%	62%	50%
Avg win	+14.8%	+26.8%	+21.3%
Avg loss	−9.2%	−1.9%	−3.0%
Crashes avoided	3	2	2

The **180-day lookback** is the clear winner:

- **+462.7% return** vs +167.5% buy & hold (**+295.2% alpha**)
- **Sharpe ratio of 1.16** — strong risk-adjusted return
- **Max drawdown of -28.1%** vs -76.6% for buy & hold — the strategy avoided the worst of the 2022 crash
- **Only in the market 34% of the time** — you hold cash two thirds of the time, dramatically reducing exposure
- **62% win rate** with average wins of +26.8% and average losses of just -1.9% — the asymmetry is excellent

Results: Ethereum (ETH-USD)

Metric	180-day
Strategy return	+86.1%
Buy & hold return	+241.4%
Alpha	−155.2%
Sharpe ratio	0.50
Max drawdown	−56.0%
Win rate	50%

The model **does not work as well on ETH**. It reduces drawdown (from -79.4% to -56.0%), but it misses too much upside. ETH's volume patterns may be noisier than BTC's (more wash trading, more DeFi-driven volume that doesn't reflect directional conviction).

Chart: BTC-USD Momentum Timing (180-day lookback)

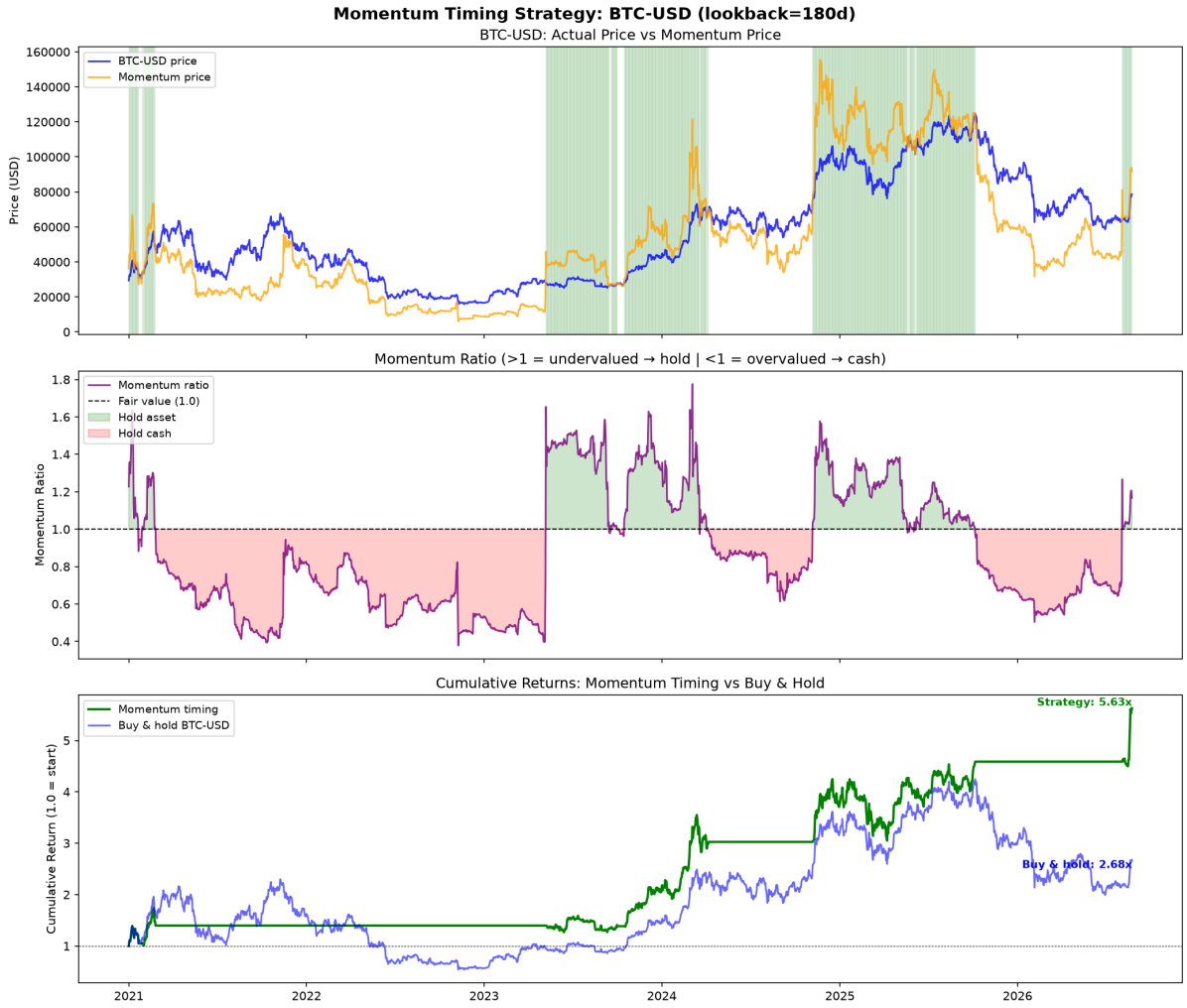


Figure 1: Top: BTC price (blue) vs momentum price (orange), with green shading when the strategy holds BTC. Middle: momentum ratio with hold/cash zones. Bottom: cumulative returns — strategy (green) vs buy & hold (blue). The strategy reaches 5.63x vs 2.68x for buy & hold.

Key observations from the chart:

1. The strategy **exits before the 2022 crash** — the momentum ratio drops below 1.0 as volume flows turn negative, signalling that the pool of money no longer supports the price.
2. It **re-enters during the 2023 accumulation phase**, when volume flows quietly turn positive while the price is still depressed.
3. It **rides the 2024–2025 bull run** but exits during corrections, preserving gains.
4. The strategy spends two thirds of its time in cash, dramatically reducing stress and exposure.

Lookback Window Sensitivity

- **90 days**: Too reactive. Generates 22 trades with a 45% win rate. Gets whipsawed by short-term volatility. Average losses (-9.2%) are too large relative to wins (+14.8%).
- **180 days**: The sweet spot. 16 trades, 62% win rate. Captures the major trend shifts (bear market entry/exit, bull market participation) without overtrading. The win/loss asymmetry (+26.8% vs -1.9%) is exceptional.
- **365 days**: Too slow. Only 10 trades. Enters late and exits late. Still beats cash (+132.3%) and dramatically reduces drawdown (-33.2% vs -76.6%), but misses the best of the moves. Returns to +132% vs buy & hold's +167%.

Interpretation: Bitcoin has a roughly 6-month momentum cycle. The 180-day window aligns with this natural rhythm — long enough to filter noise, short enough to catch regime changes. This is consistent with the ASX backtest, which also used 6-month (182-day) evaluation periods.

Caveats and Risks

1. **Backtest ≠ live trading.** These results assume you can trade at the daily close price with no slippage, no fees, and no delay. Real execution will be worse.
2. **Small sample.** 16 trades over 5 years is not statistically robust. The 62% win rate could easily be 50% or lower out of sample. The strong alpha is driven by a few large wins (the 2022 crash avoidance, the 2024 re-entry).
3. **Volume data quality.** The algorithm depends on volume reflecting genuine directional conviction. Crypto exchanges are known for wash trading (inflating volume for rankings). The yfinance data aggregates across exchanges, which helps, but the signal may be noisier than on regulated equity markets.
4. **No transaction costs.** Each trade would incur exchange fees (typically 0.1–0.5% per trade on Kraken). With 16 trades (32 transactions), this costs roughly 3–16% over the period. This reduces but does not eliminate the alpha.
5. **Tax implications.** Each exit from BTC is a taxable event. Frequent trading creates short-term capital gains, taxed at your marginal rate. Buy & hold defers tax until sale and may qualify for the 50% CGT discount (held > 12 months) in Australia.
6. **Survivor bias.** We tested BTC *because we know it went up*. We would not have tested a coin that went to zero. The strategy's apparent success on BTC may partly reflect our choice of a winning asset.
7. **Regime change.** Bitcoin's market structure is evolving (ETF inflows, institutional custody, regulatory frameworks). The volume dynamics that produced this signal in 2021–2026 may not persist.

Conclusion

The bookmaker pool model, applied as a **single-asset timing signal on Bitcoin** with a **180-day lookback window**, produced strong results in backtest: +462.7% return, +295.2% alpha over buy & hold, Sharpe ratio of 1.16, and max drawdown of -28.1% (vs -76.6% for buy & hold). The strategy is in the market only 34% of the time.

The **pairs trade** (BTC/ETH relative value) failed completely and is abandoned.

The **ETH single-asset timing** was mediocre and is not recommended.

Next Steps

1. **Paper trade.** Run the 180-day BTC signal in real-time (no real money) for 3–6 months to see if the signal quality holds out of sample.
2. **Test on other lookback windows** around 180 (150, 200, 210) to check sensitivity more finely.
3. **Add transaction costs and slippage** to the simulation for more realistic results.
4. **If paper trading confirms the signal:** open a Kraken account, deposit a small amount of BTC, and run the strategy live with real money using the CCXT Python library for execution.

— *Robin*

Source Code

Core Algorithm: Bookmaker Pool with Rolling Window

momentum_single.py --- compute_rolling_momentum()

```python
def compute_rolling_momentum(history, lookback_days):
    """
    Compute rolling momentum price using the bookmaker pool model.

    Volume is normalised (divided by window mean) to prevent overflow
    on crypto's enormous trade volumes.

    Returns DataFrame with columns: Close, MomentumPrice, MomentumRatio.
    """
    closes = history["Close"].values
    volumes = history["Volume"].values
    dates = history.index

    momentum_prices = [np.nan] * len(history)
    momentum_ratios = [np.nan] * len(history)

    for i in range(lookback_days, len(history)):
        window_close = closes[i - lookback_days:i + 1]
        window_vol = volumes[i - lookback_days:i + 1]

        if len(window_close) < 10:
            continue

        # Daily returns
        returns = np.diff(window_close) / window_close[:-1]
        vols = window_vol[1:]

        # Normalise volume to prevent overflow
        mean_vol = vols.mean()
        if mean_vol == 0:
            continue
        norm_vols = vols / mean_vol

        # Net flow = normalised_volume * return
        net_flows = norm_vols * returns

        # Run the bookmaker pool
        pool = float(window_close[0])
        mp = pool

        for j in range(len(net_flows)):
            pool += net_flows[j] * mp
            mp = pool

        momentum_prices[i] = mp
        momentum_ratios[i] = mp / closes[i]

    result = pd.DataFrame({
        "Close": closes,
        "MomentumPrice": momentum_prices,
        "MomentumRatio": momentum_ratios,
    }, index=dates)

    return result.dropna()
```

Trading Simulation

momentum_single.py --- simulate_timing()

```python
def simulate_timing(asset_symbol, lookback_days, start_date, end_date):
    """
    Simulate the timing strategy.

    Rule:
      - If MomentumRatio > 1.0: hold the asset
      - If MomentumRatio <= 1.0: hold cash

    Uses previous day's signal (can't trade on today's close).
    """
    fetch_start = start_date - timedelta(days=lookback_days + 60)

    ticker = yf.Ticker(asset_symbol)
    hist = ticker.history(start=fetch_start, end=end_date, interval="1d")

    df = compute_rolling_momentum(hist, lookback_days)

    # Trim to simulation period
    start_ts = pd.Timestamp(start_date)
    if df.index.tz is not None:
        start_ts = start_ts.tz_localize(df.index.tz)
    df = df[df.index >= start_ts]

    # Signals: 1 = hold asset, 0 = hold cash
    df["signal"] = (df["MomentumRatio"] > 1.0).astype(int)

    # Position: use previous day's signal (no lookahead)
    df["position"] = df["signal"].shift(1).fillna(0).astype(int)

    # Returns
    df["asset_return"] = df["Close"].pct_change().fillna(0)
    df["strategy_return"] = df["position"] * df["asset_return"]

    # Cumulative
    df["cumulative_asset"] = (1 + df["asset_return"]).cumprod()
    df["cumulative_strategy"] = (1 + df["strategy_return"]).cumprod()

    return df
```

Failed Pairs Trade Algorithm (for reference)

momentum_crypto.py --- pairs trade signal generation

```python
def compute_momentum_price(history):
    """
    Bookmaker model adapted for crypto (single window).
    Volume normalised to prevent overflow.
    """
    h = history.copy()
    h["DailyReturn"] = h["Close"].pct_change()

    mean_vol = h["Volume"].mean()
    if mean_vol == 0:
        return None
    h["NormVolume"] = h["Volume"] / mean_vol
    h["NetVolumeFlow"] = h["NormVolume"] * h["DailyReturn"]
```

```
    h = h.dropna()

    pool = float(h["Close"].iloc[0])
    mp = pool
    momentum_prices = [mp]

    for i in range(1, len(h)):
        net_flow = float(h["NetVolumeFlow"].iloc[i])
        pool += net_flow * mp
        mp = pool
        momentum_prices.append(mp)

    return pd.Series(momentum_prices, index=h.index,
                     name="MomentumPrice")


# Signal generation for pairs trade:
# BUY base (sell quote) when base undervalued AND quote overvalued
df["signal"] = 0
df.loc[
    (df["base_momentum_ratio"] > 1 + threshold) &
    (df["quote_momentum_ratio"] < 1 - threshold),
    "signal"
] = 1   # long base / short quote

df.loc[
    (df["base_momentum_ratio"] < 1 - threshold) &
    (df["quote_momentum_ratio"] > 1 + threshold),
    "signal"
] = -1  # short base / long quote
```

Running the Backtest

Shell commands

```
# Setup
cd ~/git/andy/crypto
python3 -m venv venv
source venv/bin/activate
pip install yfinance pandas numpy matplotlib

# Single-asset timing (the one that works)
python momentum_single.py --asset BTC-USD --lookback 180 --plot

# Pairs trade (the one that doesn't)
python momentum_crypto.py --base ETH-USD --quote BTC-USD \
       --threshold 0.01 --plot

# Try different lookback windows
python momentum_single.py --asset BTC-USD --lookback 90 --plot
python momentum_single.py --asset BTC-USD --lookback 365 --plot

# Custom date range
python momentum_single.py --asset BTC-USD --lookback 180 \
       --start 2020-01-01 --end 2026-08-24 --plot
```